

Machine Learning for Natural Language Understanding

Prof. Achim Rettinger

Sources: Lecture

- Llama 3 (Section 3)
 - <https://ai.meta.com/research/publications/the-llama-3-herd-of-models/>
- DeepSeek-V3 (Section 4)
 - <https://arxiv.org/pdf/2412.19437>
- Data Sets Overview (Section 2)
 - <https://arxiv.org/pdf/2402.18041>

Flipped Classroom Questions

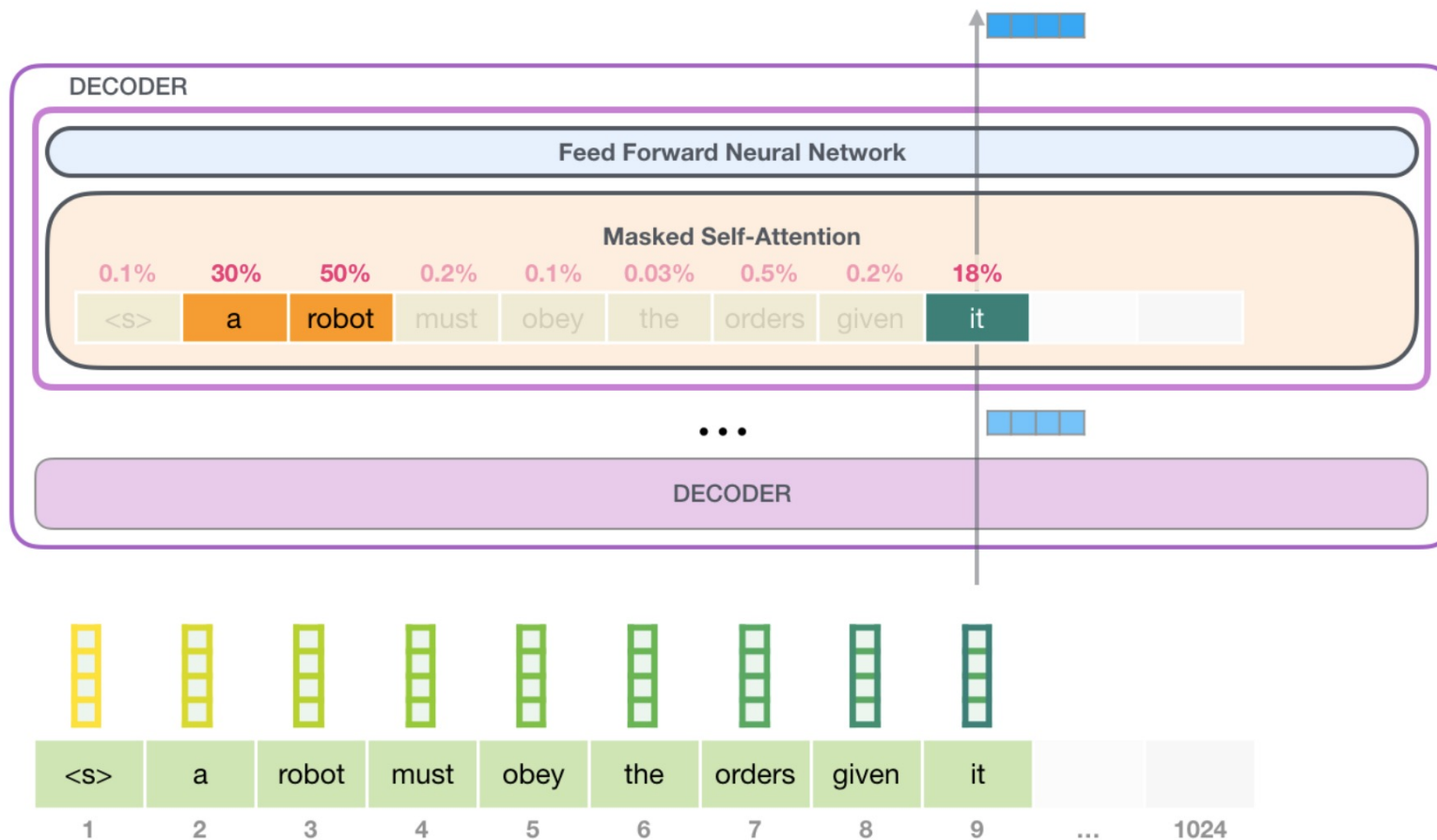
- What is the difference between training and inference?
- What is the typical pre-training objective in bidirectional encoders vs autoregressive decoders?
- Is there a way to mimic masked language modeling in autoregressive decoders?
- What is main source of data for pre-training?
- What is the context window of Llama 3?



Transformers

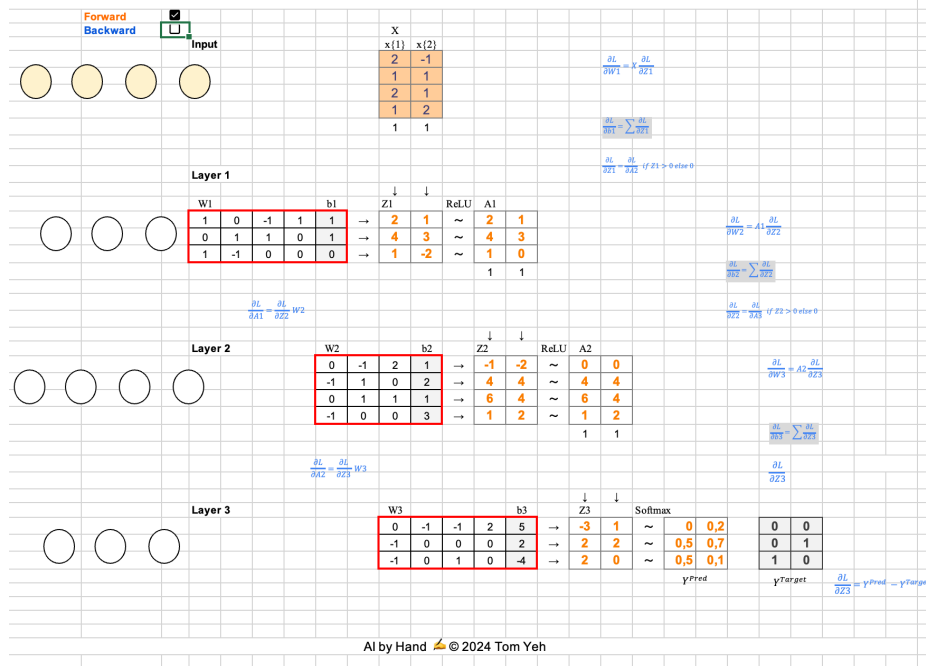
Chapter 8 – Pretraining

Recap: Next-Token Prediction

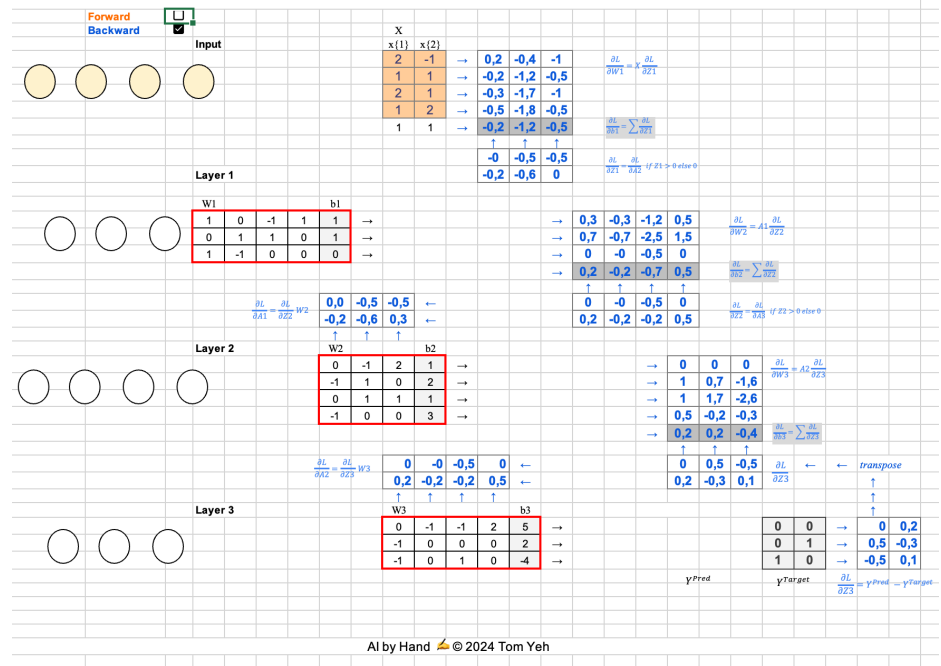


Clarification: Inference vs Training

Forward = Prediction = Inference



Backpropagation = Learning = Training

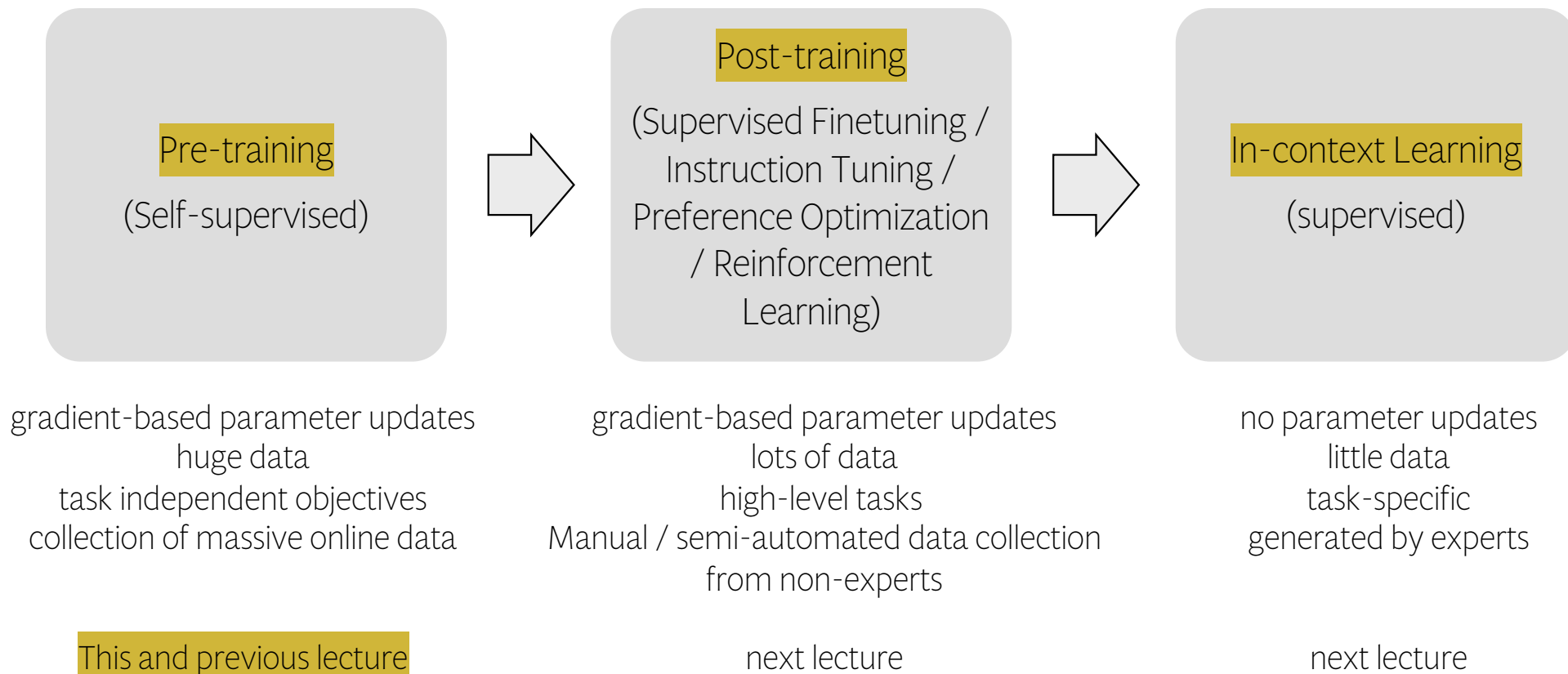


See 06-ML4TMW-Backpropagation.xlsx

Training Steps of LLMs

Chapter 8.1

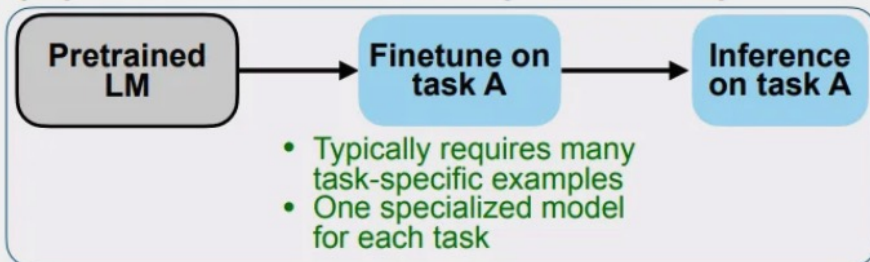
Training Steps of Instruction-tuned LLMs



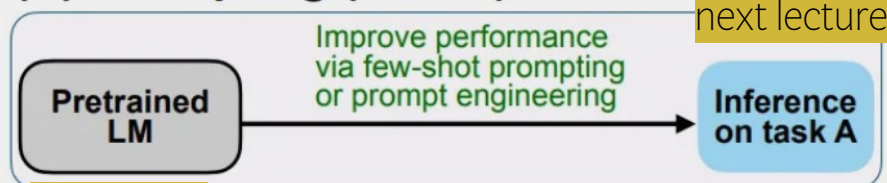
History of Training Paradigms of LLMs

this and previous lectures

(A) Pretrain–finetune (BERT, T5)



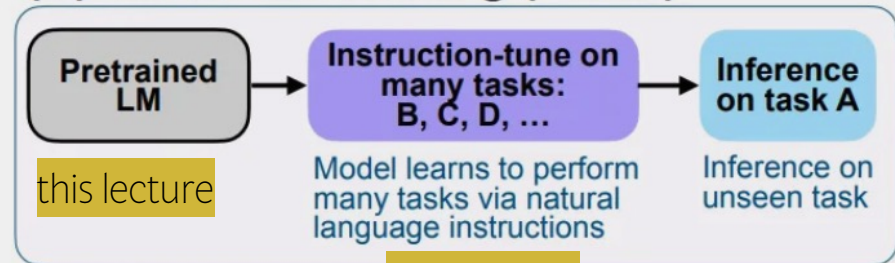
(B) Prompting (GPT-3)



this lecture

next lecture

(C) Instruction tuning (FLAN)



this lecture

next lecture

Pretrain-finetuning vs prompting vs instruction tuning (Wei et al., 2022).

Recap: Transfer Learning

- If there isn't enough labeled training data to train your network from scratch for Task 2.
- And there already exists a network that is pre-trained on a similar task (e.g., Task 1)
- And, if Task 1 and Task 2 have the same input(/output) format.
- => Use the network (with minor adaptation) for Task 2

Zero-/One-/...-/Few-Shot Learning

- No / one / ... / few training data points available for Task 2

In-Context Learning (“In-Context Inferencing”)

- Don't adapt any weights of the network during transfer

Terminology: In-Context Learning

- The LLMs weights are frozen, the task is novel to the LLM and only shown via the prompt (instructions).

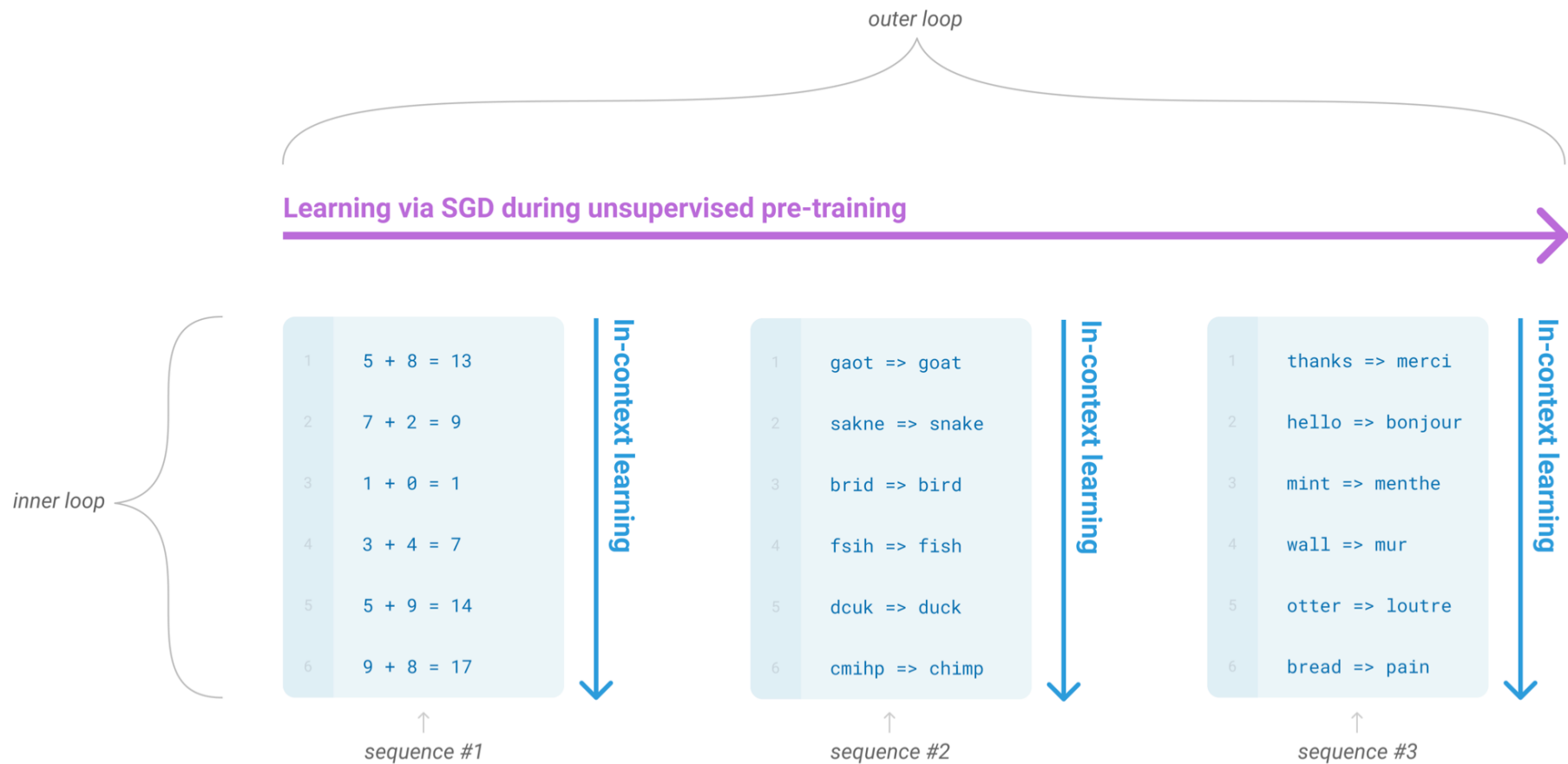
Few-Shot In-Context Learning

- The prompt includes few examples of the task

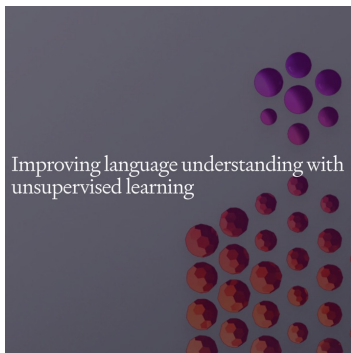
Zero-Shot In-Context Learning

- No examples, just a description of the task

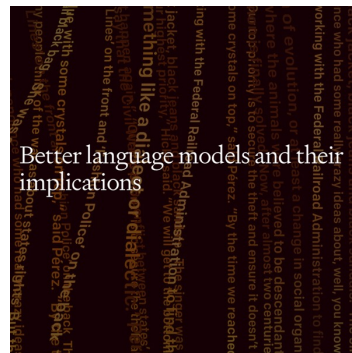
In-Context Learning (Training Paradigm B) (<https://arxiv.org/pdf/2005.14165.pdf>)



Evolution of GPT



- 2018
- 12 layers x 12 heads
- Input context: 512
- Zero-shot performance



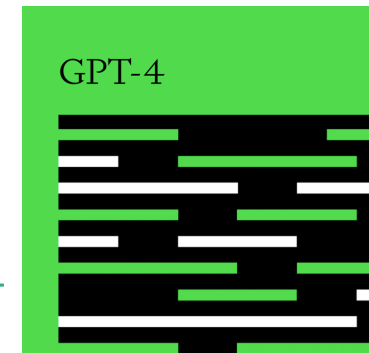
- 2019
- 48 layers x 48 heads
- Input context : 1024
- Task conditioning

Language models are few-shot learners

[Read paper](#) [Language GPT-3 Publication Milestone](#)

Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question answering, and short tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss



- 2020
- 96 layers x 96 attention
- Input context: 2048
- In-context learning

Training Paradigm B

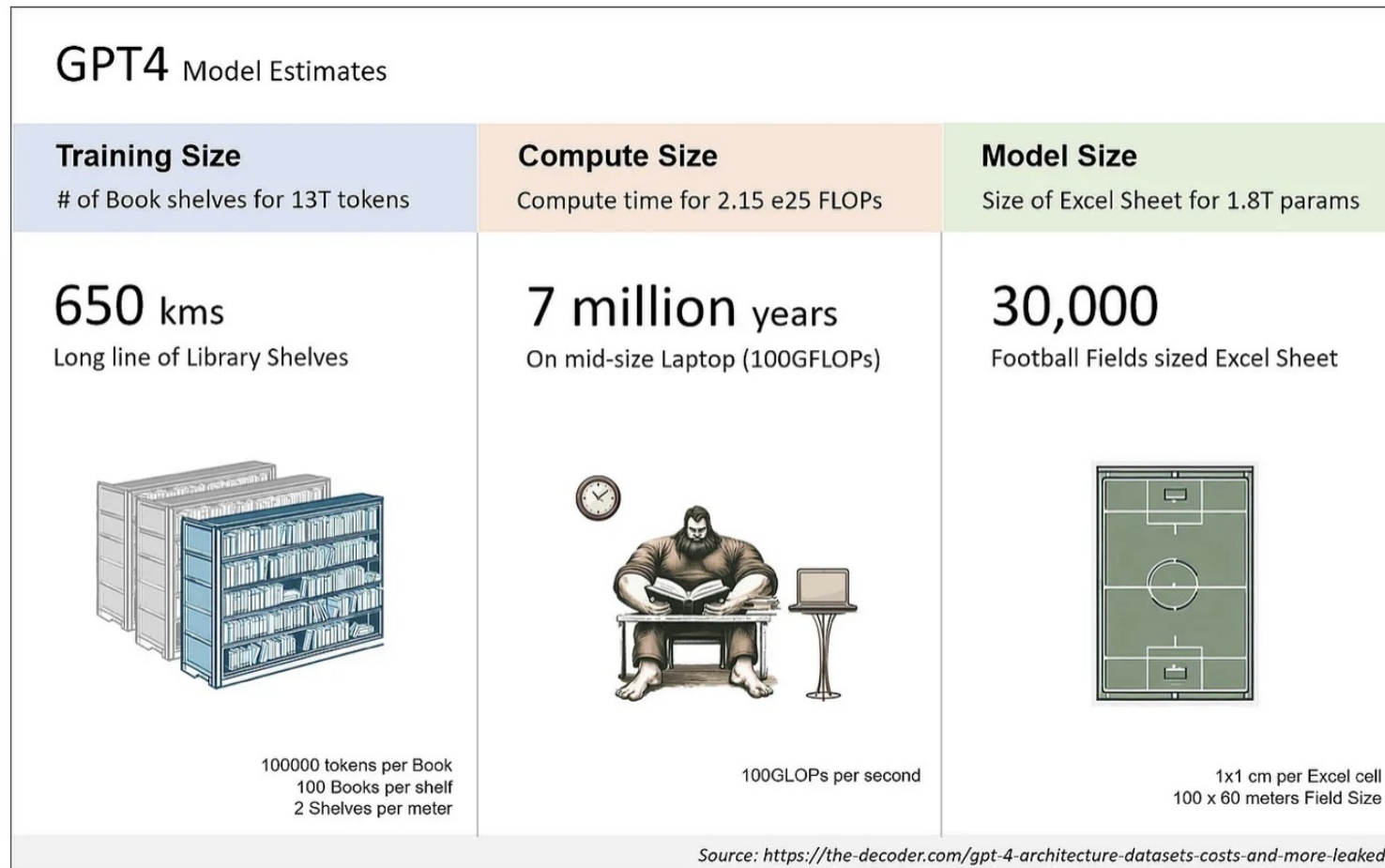
- 2022
- Instruct GPT / ChatGPT

- Instruction-following LLM

Training Paradigm C

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Pre-training Data Set Size

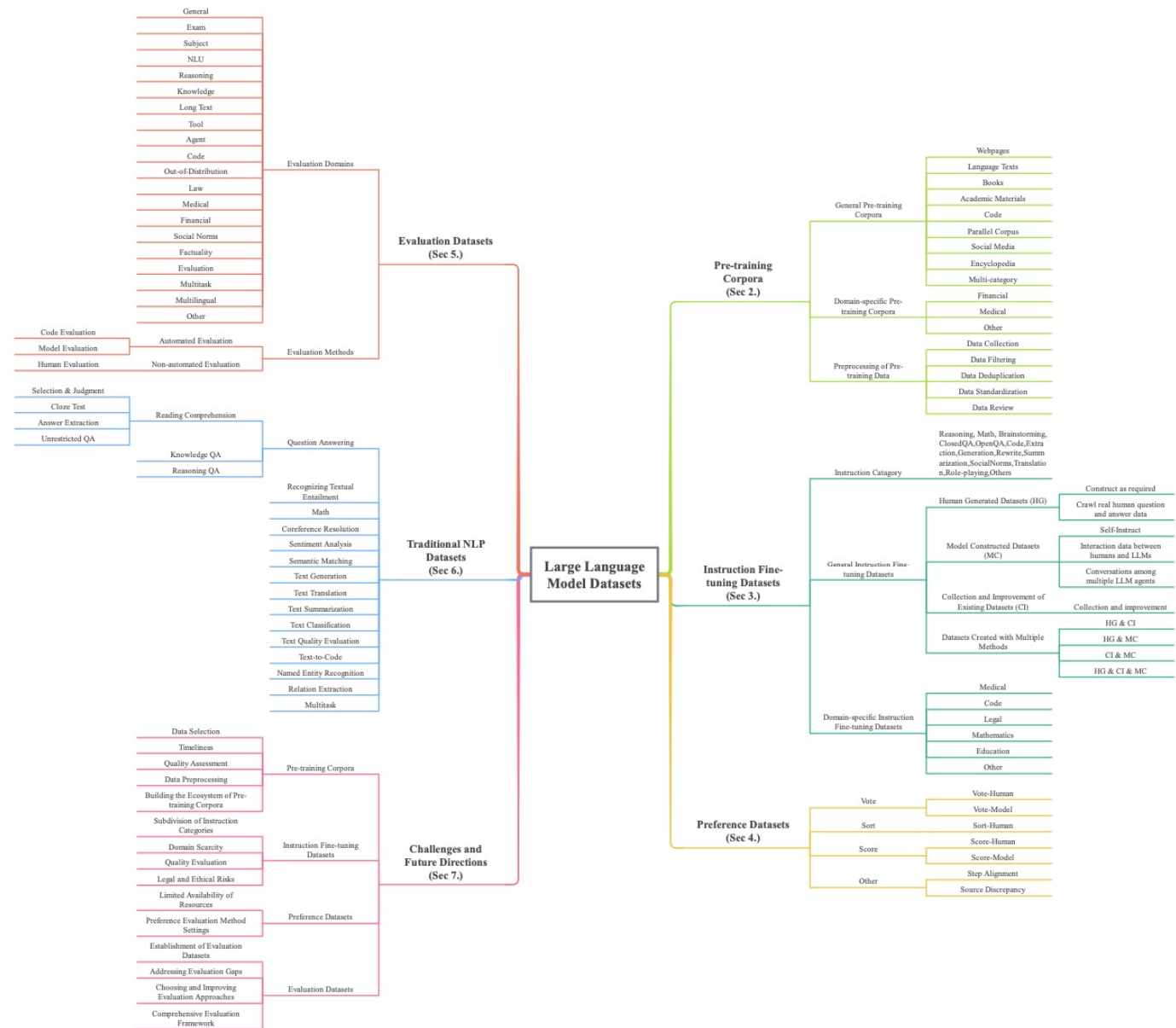


Data for Pretraining LLMs

Chapter 8.2

Data Sets Overview

(<https://arxiv.org/pdf/2402.18041>)



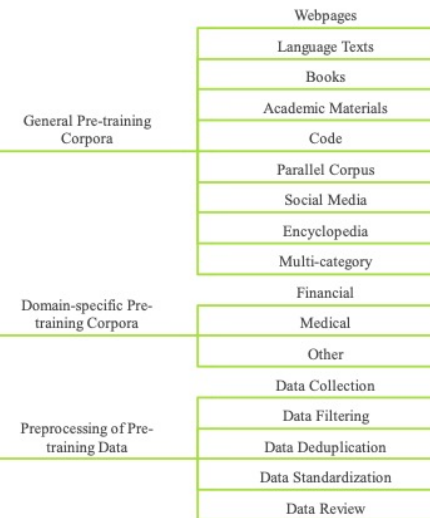
Data Sets Overview

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Large Language Model Datasets

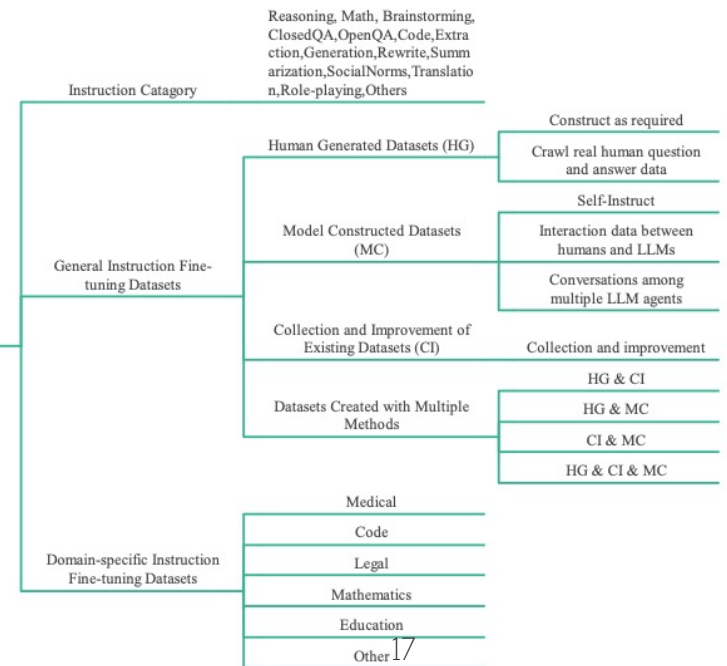
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Pre-training Corpora (Sec 2.)



next lecture

Instruction Fine-tuning Datasets (Sec 3.)

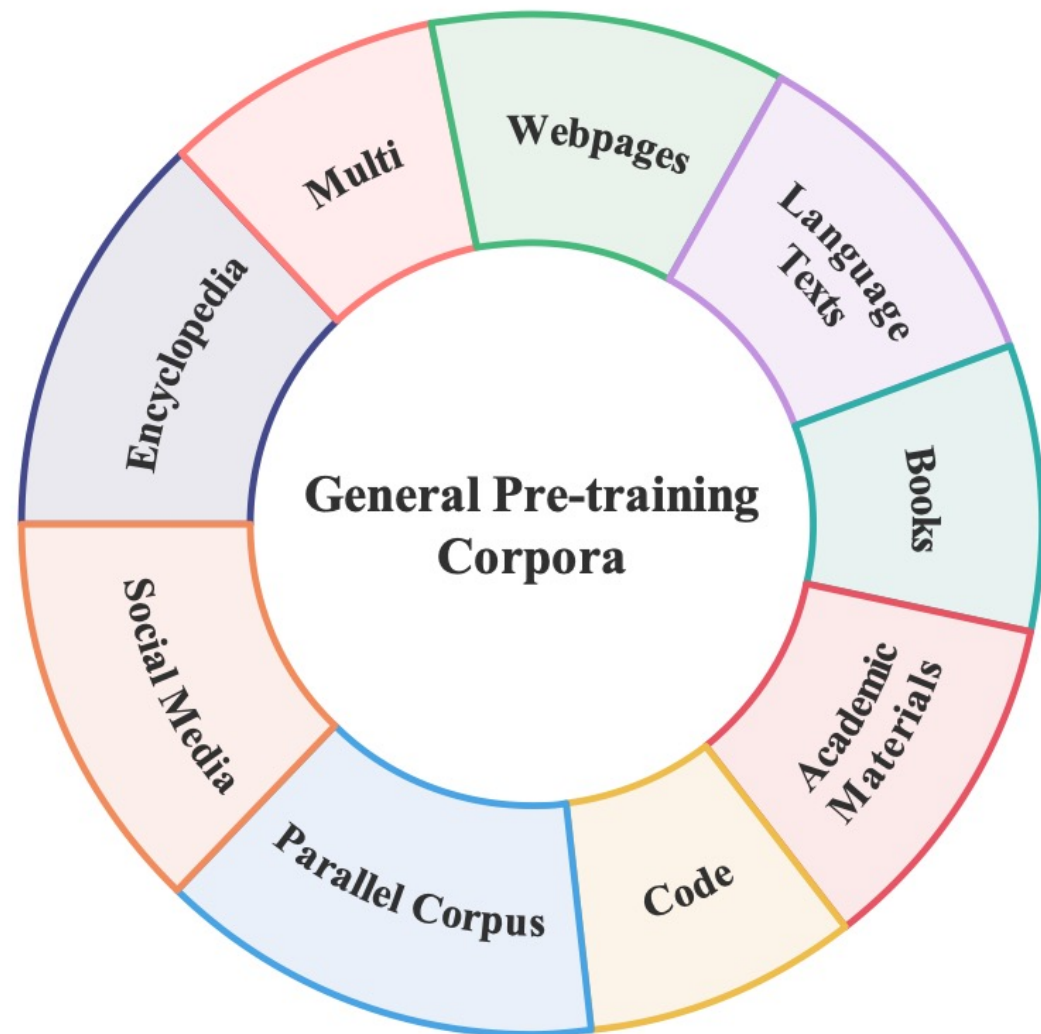


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Types of Data of General Pre-training Corpora

(<https://arxiv.org/pdf/2402.18041>)

Largest Source:
<https://commoncrawl.org>



Pre-Training Data

Characteristics

- Massive Scale
- Multilingual
- Broad Topics / Content
- Semi-structured
- Messy
- Contains illegal / sensitive / explicit / biased content
- Requires extensive cleaning

Function

- Provides General Linguistic Capabilities
 - Mostly: Syntax, Morphology, Semantics
 - Some: Pragmatics (more in post-training)
- Provides (multi-) Language Specific Capabilities
- Provides General Knowledge

Web Data Curation (Llama 3)

Filtering

- Personally identifiable information filtering
- Safety filtering
- HTML cleaning
- De-duplication
- Heuristic Filtering
- Model-based quality filtering (using e.g., Llama 2)

Categorization

- Math deduction
- Reasoning in STEM areas
- Code
- Multilingual data

“Final data mix contains roughly 50% of tokens corresponding to general knowledge, 25% of mathematical and reasoning tokens, 17% code tokens, and 8% multilingual tokens.”

Selected Pre-Training-Process Details

Chapter 8.3

Selected Pre-Training Details (Llama 3 405B)

- Trained on 16K H100 GPUs, 80GB memory
- Pre-trained using AdamW
- Increasing batch size during training
 - Initial batch size of 4M tokens and sequences of length 4,096
 - batch size of 8M sequences of 8,192 tokens after pre-training 252M tokens
 - double batch size again to 16M after pre-training on 2.87T tokens
- Increasing context window during training
 - Gradually in six stages
 - starting from 8K context window
 - ending with a final 128K context window

Selected Pre-Training Details (DeepSeek-V3)

- training corpus consists of 14.8T high-quality and diverse tokens
- data augmentation by using the Fill-in-Middle (FIM) strategy to emulate masked language modeling.
- BPE with an extended vocabulary of 128K tokens
- number of Transformer layers 61, embedding dimension 7168
- substitute all FFNs with MoE layers: 1 shared expert and 256 routed experts, of which only 8 experts will be activated for each token
- 671B total parameters but only 37B are activated for each token
- AdamW optimizer
- Long Context Extension: 4K to 32K and then to 128K

Evaluation of Pretraining

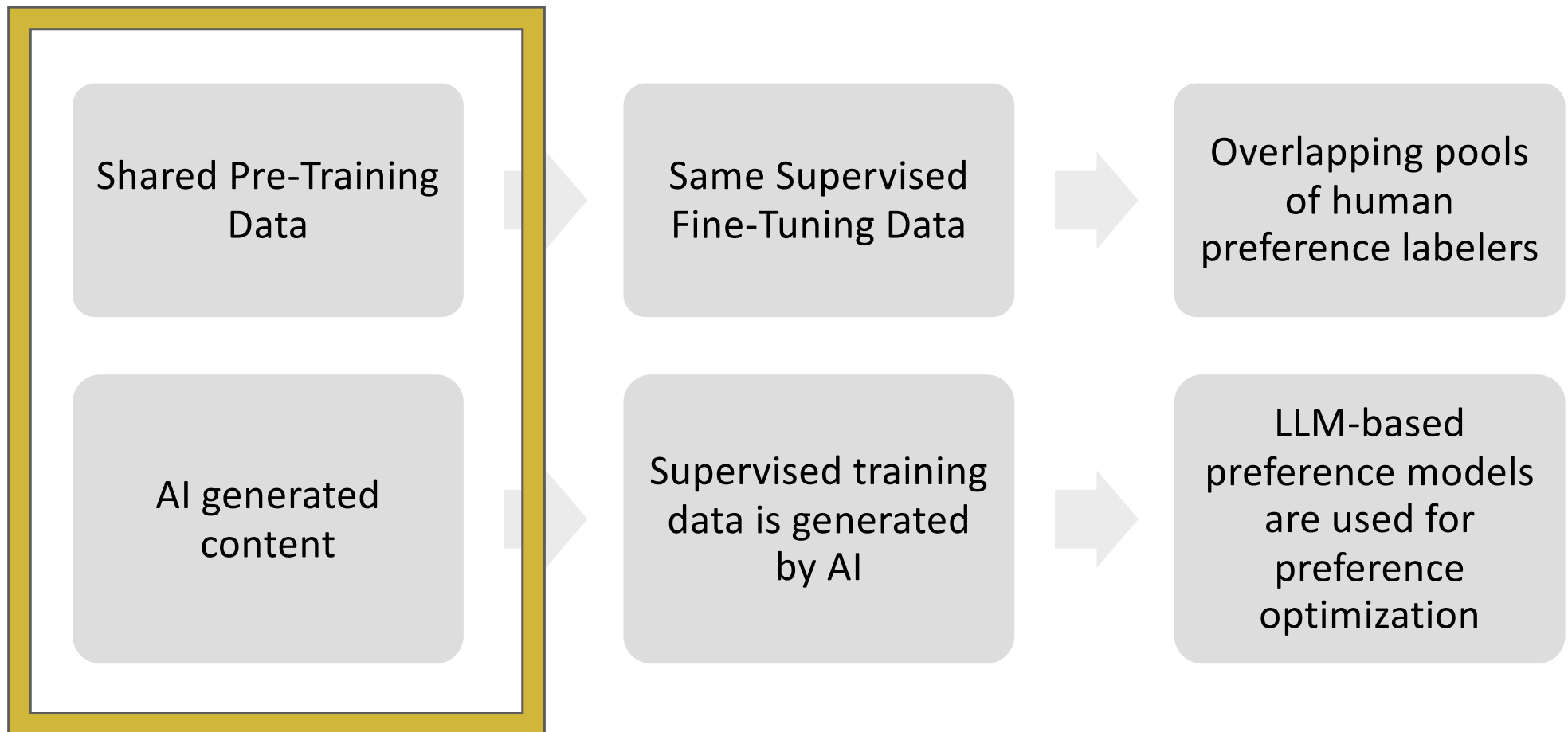
Chapter 8.2

Pre-Training Benchmarks

<https://arxiv.org/pdf/2412.19437>

	Benchmark (Metric)	# Shots	DeepSeek-V2 Base	Qwen2.5 72B Base	LLaMA-3.1 405B Base	DeepSeek-V3 Base
	Architecture	-	MoE	Dense	Dense	MoE
	# Activated Params	-	21B	72B	405B	37B
	# Total Params	-	236B	72B	405B	671B
English	Pile-test (BPB)	-	0.606	0.638	0.542	0.548
	BBH (EM)	3-shot	78.8	79.8	82.9	87.5
	MMLU (EM)	5-shot	78.4	85.0	84.4	87.1
	MMLU-Redux (EM)	5-shot	75.6	83.2	81.3	86.2
	MMLU-Pro (EM)	5-shot	51.4	58.3	52.8	64.4
	DROP (F1)	3-shot	80.4	80.6	86.0	89.0
	ARC-Easy (EM)	25-shot	97.6	98.4	98.4	98.9
	ARC-Challenge (EM)	25-shot	92.2	94.5	95.3	95.3
	HellaSwag (EM)	10-shot	87.1	84.8	89.2	88.9
	PIQA (EM)	0-shot	83.9	82.6	85.9	84.7
	WinoGrande (EM)	5-shot	86.3	82.3	85.2	84.9
	RACE-Middle (EM)	5-shot	73.1	68.1	74.2	67.1
	RACE-High (EM)	5-shot	52.6	50.3	56.8	51.3
	TriviaQA (EM)	5-shot	80.0	71.9	82.7	82.9
	NaturalQuestions (EM)	5-shot	38.6	33.2	41.5	40.0
	AGIEval (EM)	0-shot	57.5	75.8	60.6	79.6
Code	HumanEval (Pass@1)	0-shot	43.3	53.0	54.9	65.2
	MBPP (Pass@1)	3-shot	65.0	72.6	68.4	75.4
	LiveCodeBench-Base (Pass@1)	3-shot	11.6	12.9	15.5	19.4
	CRUXEval-I (EM)	2-shot	52.5	59.1	58.5	67.3
	CRUXEval-O (EM)	2-shot	49.8	59.9	59.9	69.8
Math	GSM8K (EM)	8-shot	81.6	88.3	83.5	89.3
	MATH (EM)	4-shot	43.4	54.4	49.0	61.6
	MGSM (EM)	8-shot	63.6	76.2	69.9	79.8
	CMath (EM)	3-shot	78.7	84.5	77.3	90.7
Chinese	CLUEWSC (EM)	5-shot	82.0	82.5	83.0	82.7
	C-Eval (EM)	5-shot	81.4	89.2	72.5	90.1
	CMMLU (EM)	5-shot	84.0	89.5	73.7	88.8
	CMRC (EM)	1-shot	77.4	75.8	76.0	76.3
	C3 (EM)	0-shot	77.4	76.7	79.7	78.6
	CCPM (EM)	0-shot	93.0	88.5	78.6	92.0
Multilingual	Arabic MMLU-non-English (EM)	5-shot	64.0	74.8	73.8	79.4

Cross-Model Homogeneity and Data Poisoning



"Whoever controls (the curation of) the training data and the finetuning, defines the cultural identity of AI models and ultimately of the communities who use them. In democracies this should make those models a public good and cultural communities should have control over the curation of the corpora and the methodologies and policies for the finetuning of the models."

“The future of language is about the development of language models...”¹



1

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